

* RECAP

- Throughput depends on bottleneck bandwidth & delays

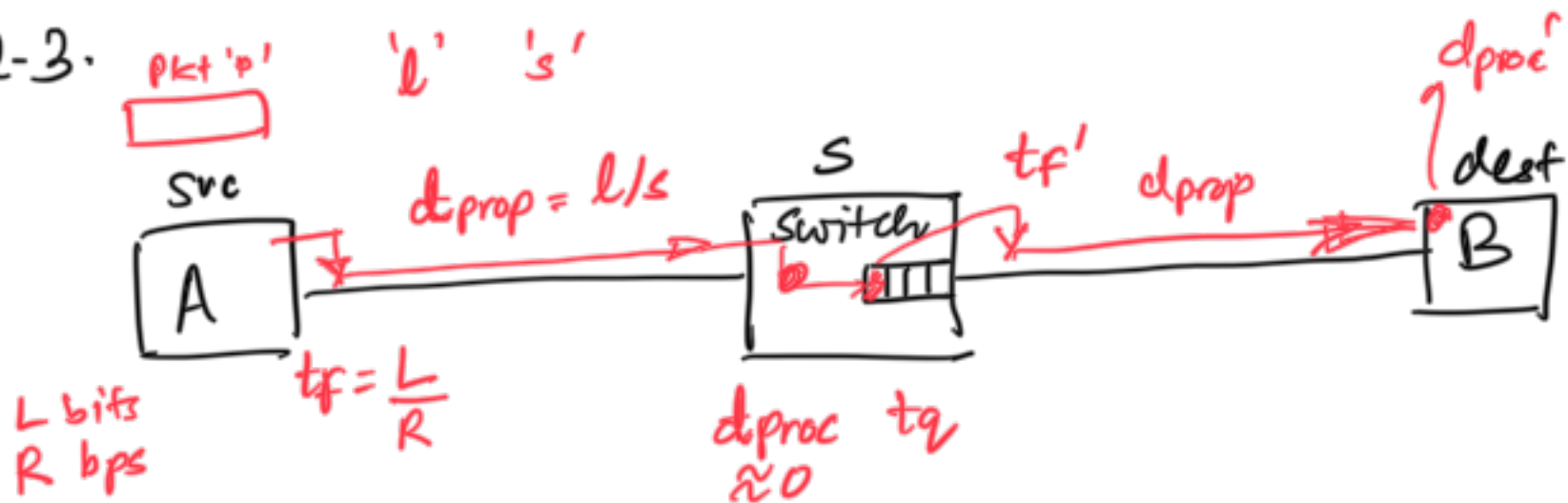
- Delays could occur due to many reasons: protocol design (wait for ACKs, ...), packet losses, retransmissions, nodal delays (processing, queuing, ...), propagation delay

* DELAYS

- Transmission delay (t_{trans} or t_f): time reqd to place all the bits of a packet (data) from the sender's buffer on the link
 - Link data rate = R bits/sec
 - Packet size = L bits
 - Transmission delay, $t_{trans} = \frac{L}{R}$ seconds
- Propagation delay (t_{prop}): time reqd for one bit to propagate from one end to another.
 - Length of the link / Distance to be travelled = l meters
 - Speed at which signal travels = s meters/sec
 - Propagation delay, $t_{prop} = \frac{l}{s}$ seconds
- Processing delay (t_{proc}): time involved in
 - lookup into routing / switching table
 - computation such as header checksum, ...
 - Not significant, due to high speed hardware
- Queuing delay (t_q): packet may be queued until the next link becomes free

End-to-end delays

Q-3.



$$d_{\text{end-to-end}} = t_f + d_{prop_{A-B}} + d_{proc_S} + t_q + t_f' + d_{prop_{S-B}} + d_{proc_B}$$

$$L = 125 \text{ B} = 125 * 8 = 1000 \text{ bits}$$

$$R = 10^6 \text{ bps}$$

$$t_f = \frac{1000}{10^6} = 1000 \mu\text{s}$$

$$l = 200 \text{ m}$$

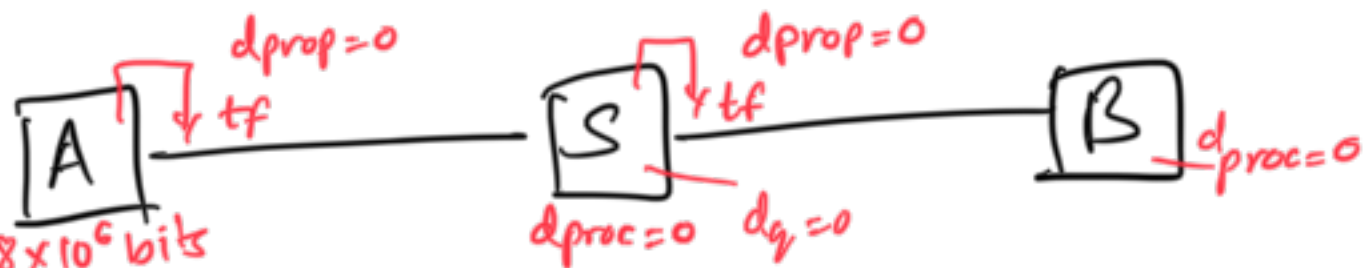
$$s = 2 \times 10^8 \text{ m/s}$$

$$d_{prop} = \frac{200}{2 \times 10^8} = 1 \mu\text{s}$$

$$d_{\text{end-to-end}} = 1000 + 1 + 9 + 0 + 1000 + 1 + 9$$

$$= 2020 \mu\text{s}$$

Q.4. (a)



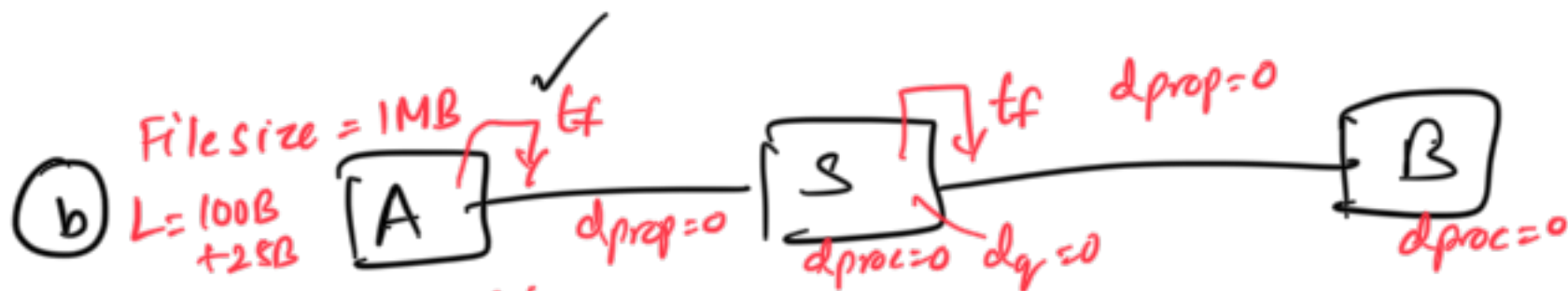
$$L = 1\text{MB} = 8 \times 10^6 \text{ bits}$$

$$R = 1\text{Mbps}$$

$$t_f = \frac{L}{R} = 8\text{sec}$$

$$d_{\text{tot}} = 8 + 8$$

$$\underline{d_{\text{tot}} = 16\text{sec}}$$



$$L = 125\text{B} = 125 \times 8 = 1000\text{bits}$$

$$\# \text{pkts} = \frac{1\text{MB}}{100\text{B}} = 10,000$$

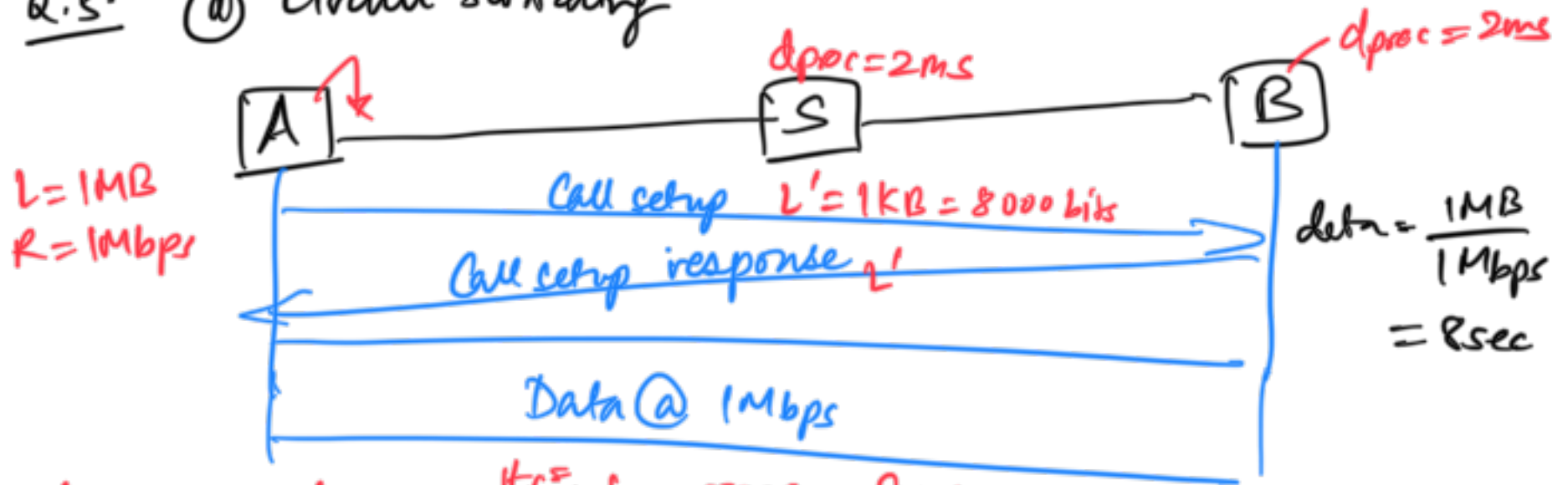
$$t_f = \frac{1000}{10^6} = \underline{1\text{msec}}$$

$$t_{fA} = t_{fS}$$

$$d_{\text{tot}} = t_{fA} \times 10000 + \cancel{d_{\text{prop}}} + \cancel{d_{\text{proc}}} + t_{fS} + \cancel{d_{\text{prop}}} + \cancel{d_{\text{proc}}}$$

$$\underline{d_{\text{tot}} = 10.001\text{sec}}$$

Q.5. (a) Circuit switching



$$d_{call setup} = d_{call resp.} \quad t_f = \frac{L'}{R} = \frac{8000}{10^6} = 8ms$$

$$= (t_f + d_{proc}) * 2$$

$$= (8 + 2) * 2$$

$$= 20ms$$

$$d_{total} = d_{call setup} + d_{call resp} + d_{data}$$

$$= 20 + 20 + 8sec$$

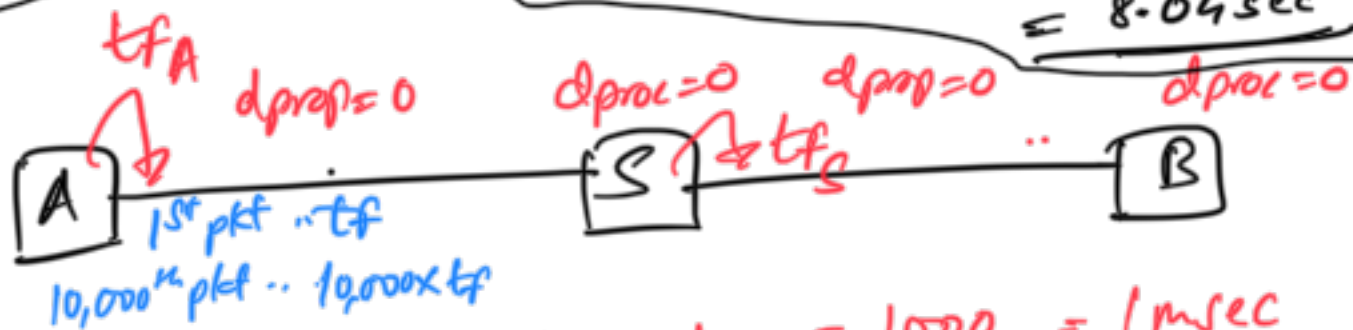
$$= 8.04sec$$

(b) Packet switching

$R = 1Mbps$
 $L = 1MB$

$$L' = 100B + 25B = 1000 \text{ bits}$$

$$\# \text{ pkts} = \frac{1MB}{100B} = 10,000$$



$$t_{fA} = t_{fS} = \frac{1000}{10^6} = 1msec$$

$$d_{total} = t_{fA} * 1000 + \cancel{d_{prop}} + \cancel{d_{proc}} + t_{fS} + \cancel{d_{prop}} + \cancel{d_{proc}}$$

only for the last pkt

$$= 1msec * \overset{\text{num-pkts}}{10000} + 1msec$$

$$= 10sec + 1msec$$

$$d_{total} = 10.001sec$$

	delay	Total bytes sent	Overhead
Circuit switching	8.04 sec	1MB + 1KB + 1KB = 1.002MB	2KB
Packet switching	10.001 sec	10,000 * 125B = 1.25 MB	250KB

